

Generative AI & Agentic AI — Interview Questions

50 Must-Know Questions for 2025–2026 | No answers — test yourself!

SECTION 1: GENERATIVE AI — FUNDAMENTALS

Basic / Conceptual Questions

1. What is Generative AI? How does it differ from Traditional AI?
 2. What are the most common types of Generative AI models?
 3. What is a Transformer and why is it fundamental to Generative AI?
 4. Explain the difference between a Generative model and a Discriminative model.
 5. What is the role of Tokens and Embeddings in LLMs?
 6. What is Attention Mechanism and how does it work?
 7. What is the difference between GPT and BERT?
 8. Explain how GANs work. What are their training challenges?
 9. What are VAEs (Variational Autoencoders)? How do they differ from GANs?
 10. What are Diffusion Models and how do they generate images?
 11. What is Hallucination in LLMs and how can it be reduced?
 12. What is the context window in LLMs? Why does it matter?
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Intermediate — Fine-tuning, RAG & Prompt Engineering

13. What is Retrieval-Augmented Generation (RAG)? Explain its architecture.
 14. When would you choose RAG vs Fine-tuning?
 15. Explain LoRA and QLoRA. Why are they used for fine-tuning LLMs?
 16. What is Prompt Engineering? List important techniques.
 17. What is the difference between Zero-Shot, One-Shot, and Few-Shot learning?
 18. What are Vector Databases? Which ones are commonly used?
 19. How do you evaluate the performance of a Generative AI model?
 20. What are chunking strategies in RAG? How do you choose chunk size?
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Advanced — Architecture, Safety & System Design

21. What is RLHF (Reinforcement Learning from Human Feedback)?
22. What is the difference between Instruction Tuning and RLHF?

23. What are some key ethical concerns in Generative AI?
 24. How would you optimize inference speed for a large GenAI model in production?
 25. What are Multimodal Models? Give examples.
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SECTION 2: AGENTIC AI — ALL LEVELS

Fundamentals of Agentic AI

26. What is Agentic AI? How does it differ from a standard LLM chatbot?
 27. What are the core components of an AI Agent?
 28. What are the different types of AI agents?
 29. What is the ReAct (Reasoning + Acting) framework?
 30. What is Chain-of-Thought (CoT) reasoning and why is it important for agents?
 31. Explain the difference between Single-Agent and Multi-Agent systems.
 32. What is memory management in AI agents? Explain types of memory.
 33. What is Tool Use / Function Calling in LLM agents?
 34. What is the Model Context Protocol (MCP)?
 35. What are common Agentic AI design patterns?
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Advanced — Architecture, Safety & Production

36. What is Agentic RAG and how does it differ from standard RAG?
 37. What is LangChain? How does it compare to LlamaIndex and LangGraph?
 38. What are the security risks of autonomous AI agents and how do you mitigate them?
 39. How do you evaluate and monitor an Agentic AI system in production?
 40. What is GraphRAG and when would you use it over standard vector RAG?
 41. What is the difference between LangGraph and AutoGen for multi-agent systems?
 42. How would you design a production-grade Agentic AI system architecture?
 43. What is Human-in-the-Loop (HITL) in Agentic AI? When should you require it?
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SECTION 3: SCENARIO-BASED & SYSTEM DESIGN QUESTIONS

44. Your RAG system is hallucinating despite providing correct context. How do you debug and fix it?

45. Design a customer support agent that handles complex multi-step inquiries.
46. How would you prevent prompt injection attacks in an agentic system?
47. You need to build a coding agent (like Devin). What components would you include?
48. Describe a time you built or worked on a RAG or LLM-powered system. What challenges did you face?
49. How do you stay current with the fast-moving GenAI and Agentic AI field?
50. What are the current limitations of Agentic AI systems?

Good luck with your interview!